INTERSECTION PRODUCT OF TWO INTEGRAL SUBSCHEMES OF A SMOOTH INTEGRAL SCHEME EQUALS THE FUNDAMENTAL CLASS OF THEIR INTERSECTION UNDER A DIMENSION AND COHEN-MACAULAY HYPOTHESIS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFD (sources/stacks-chow/pinned-original-source.tex line 14523).

Allowed standard prerequisites (editorial).

- The Gysin fundamental class identity for a virtual normal sheaf under dimension and Cohen-Macaulay hypotheses. Same chapter, Lemma labelled lemma-gysin-easy (source line 12128), invoked at proof lines 14547 and 14558-14559. Supplies the conclusion once the four hypotheses about the diagonal are verified; it is counted separately in this lane as S12-algebra-gysin-easy.
- Higher codimension Gysin maps and the identification of the intersection product with a diagonal pullback. Same chapter, Section labelled section-gysin-higher-codimension (source line 11833), invoked at proof lines 14537-14540. Supplies that [Y] . [Z] = Delta^!([Y x Z]) with the Gysin map taken with respect to the locally free tangent sheaf.
- Cohen-Macaulayness goes up along a flat local homomorphism with Cohen-Macaulay fibres. Algebra, Lemma labelled algebra-lemma-CM-goes-up (Stacks tag 045J), invoked at proof lines 14556-14557. Supplies that O_{Y x Z,xi} is Cohen-Macaulay from the Cohen-Macaulayness of O_{Y,xi} and the fibre O_{Z,xi}.

Source reference. Original section 0FBI: sources/stacks-chow/pinned-original-source.tex, source lines 11833–11833. Same-chapter pointer into the author environment lemma at source line 11833; invoked at proof line 14540 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FEA: sources/stacks-chow/pinned-original-source.tex, source lines 12128–12128. Same-chapter pointer into the author environment lemma

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at source line 12128; invoked at proof line 14547 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 045J: sources/stacks-chow/tags.txt, source lines 5438–5438. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 14557 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let X be an integral scheme smooth over k. Let $Y, Z \subset X$ be integral closed subschemes. Set $d = \dim(Y) + \dim(Z) - \dim(X)$. Assume*

(1) $\dim(Y \cap Z) \leq d$, *and*
(2) $\mathcal{O}_{Y,\xi}$ *and* $\mathcal{O}_{Z,\xi}$ *are Cohen-Macaulay for every* $\xi \in Y \cap Z$ *with* $\delta(\xi) = d$.

Then $[Y] \cdot [Z] = [Y \cap Z]_d$ *in* $\mathrm{CH}_d(X)$.

Proof. Recall that $[Y] \cdot [Z] = \Delta^!([Y \times Z])$ where $\Delta^! = c(\Delta : X \to X \times X, \mathcal{T}_{X/k})$ is a higher codimension Gysin map (Section 0FBI) with $\mathcal{T}_{X/k} = \mathcal{H}om(\Omega_{X/k}, \mathcal{O}_X)$ locally free of rank $\dim(X)$. We have the equality of schemes

$$Y \cap Z = X \times_{\Delta,(X\times X)} (Y \times Z)$$

and $\dim(Y \times Z) = \dim(Y) + \dim(Z)$ and hence conditions (1), (2), and (3) of Lemma 0FEA hold. Finally, if $\xi \in Y \cap Z$, then we have a flat local homomorphism

$$\mathcal{O}_{Y,\xi} \longrightarrow \mathcal{O}_{Y\times Z,\xi}$$

whose "fibre" is $\mathcal{O}_{Z,\xi}$. It follows that if both $\mathcal{O}_{Y,\xi}$ and $\mathcal{O}_{Z,\xi}$ are Cohen-Macaulay, then so is $\mathcal{O}_{Y\times Z,\xi}$, see Algebra, Lemma 045J. In this way we see that all the hypotheses of Lemma 0FEA are satisfied and we conclude. □